

MATH 174 - Numerical Analysis

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Fall 2025

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1 Error

1.1 Approximation

Let x be an exact number.

Let y be an approximation of x .

Def: The absolute error is $|x - y|$.

Def: The relative error is $\frac{|x - y|}{|x|}$,

1.2 Round-off Error

Usually a machine uses a finite amount of memory to store a number.

machine $\leftarrow \pi$

Suppose a machine can store some number of digits.

1. Allocate a digit for sign, positive or negative.
2. Allocate a number of digits for the mantissa.
3. Allocate a digit for sign of the exponent.
4. Allocate a number of digits for the exponent.

Under rounding in the case of a 4 digit mantissa, $\pi = 0.3142 \cdot 10^1$, this is the machine # representation of pi.

Def: Round-off error are all errors arising from the error between actual # and machine #.

Notation:

$$\begin{aligned}x &= \text{real \#} \\ fl(x) &= \text{machine \# representation of } x\end{aligned}$$

In an s-digit rounding machine,

$$\text{Abs error} = |x - fl(x)|$$

$$\text{Rel error} = \frac{|x - fl(x)|}{|x|} \leq \frac{1}{2} \cdot 10^{1-5} = 5 \cdot 10^{-5}$$

1.3 Floating Point Arithmetic

What happens under arithmetic operations? (+, -, ·, /)

Two things to watch out for:

1. Many operations can degrade error.
2. Catastrophic cancellation.

1.3.1 Degrading Error

number	machine
x	$fl(x)$
y	$fl(y)$
$x + y$	$fl(fl(x) + fl(y))$

The more nested $fl()$ there are, the larger the error will be. The same follows for $-$, $\cdot$, $/$.

When performing arithmetic, it temporarily allows for more digits than the mantissa allows.

Algorithmic "solutions"

$$((x + y) + z) + w$$

Worst case, 4 times the error.

$$(x + y) + (z + w)$$

Worst case, 3 times the error.

1.3.2 Catastrophic Cancellation

Ex: 3-digit rounding machine. Given 2 numbers: 0.312 and 0.311.

$$0.312 - 0.311 = 0.001 = 0.1 \cdot 10^{-2}$$

Loss of significant digits!

Ex: 5-digit rounding machine. Given 2 numbers:

$$x = 0.41626324$$

$$y = 0.41626323$$

$$x - y = 0.00000001$$

In machine:

$$fl(x) = 0.41626$$

$$fl(y) = 0.41626$$

$$fl(x) - fl(y) = 0.00000 = 0$$

$$fl(fl(x) - fl(y)) = 0$$

Relative error:

$$\begin{aligned} \frac{|x - y - fl(fl(x) - fl(y))|}{|x - y|} &= \frac{|x - y - 0|}{|x - y|} \\ &= 1 = 100\% \text{ error!} \end{aligned}$$

2 Speed

2.1 Dot Product

Consider dot product of vectors

$$\begin{aligned} \vec{x} &\in \mathbb{R}^n, \vec{y} \in \mathbb{R}^n \\ \vec{x} \cdot \vec{y} &= \begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \\ &= x_1 y_1 + x_2 y_2 + \dots + x_n y_n \end{aligned}$$

Count time-consuming operations:

1. Add/subtract
2. Multiply/divide
3. Comparisons
4. Memory access/storage

These are called FLOPS (floating point operations).

For the dot product, there are n multiplications and $n-1$ additions.

Adding these together, we get $2n - 1$ FLOPS.

Note that this expression is linear in n .

For large n , $2n$ dominates, $= \mathcal{O}(n) \approx \text{constant times } n$.

ex: For n vs $2n$, $2n$ will take twice as long.

ex: For $\mathcal{O}(n^2)$, n vs $2n$, $2n$ will take four times as long.

2.2 Memory

Consider a matrix $A_{n \times n}$

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} vs \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

The square matrix uses n^2 memory locations, while the vector uses n memory locations. Each number also uses 8 bytes, creating another goal for algorithm development, reduce memory usage.

2.3 MATLAB

A simple MATLAB function to compute the dot product of two vectors and return the number of FLOPS used.

```
function [thedotproduct, theflops] = dotprod(x, y, n)
    thedotproduct = x(1)*y(1);
    theflops = 1;
    for i = 2:n
        thedotproduct = thedotproduct + x(i)*y(i);
        theflops = theflops + 2;
    end
end
```

2.3.1 First Problem

Write a MATLAB function that when given $A\vec{x} = \vec{b}$, computes $\vec{x}$

$$A\vec{x} = \vec{b}$$

A is triangular and invertible